TEXAS
The University of Texas at Austin

The University of Texas at Austin
Biomedical Engineering
Cockrell School of Engineering

Assessing changes in 4DCT-Derived Lung Compliance for Idiopathic Pulmonary Fibrosis (IPF) Patients using linear regression

BME 377 Semester Presentation

Hsu-Ting Kuo

Current challenges in monitoring Idiopathic Pulmonary Fibrosis (IPF)

- **IPF**: Thickens and stiffens lung tissue ➔ **Irreversible** lung scarring ➔ alters **lung compliance**
- Suffocates patients to death within **3-5** years after diagnosis
- Treatments only work for a **subset of patients** and **have significant side effects**

- **Lung compliance**: measures the **distensibility** of the lung

$$\boxed{\mathrm{LC} = \frac{\Delta V}{\Delta P}}$$

Mechanical ventilation Commonly used	Intrapleural pressure measurement	Esophageal manometry	Four-dimensional CT (4DCT) Least invasive
Invasive	Invasive	Invasive	Non-invasive

CT scanning that records multiple images over **time**

Current challenges in monitoring Idiopathic Pulmonary Fibrosis (IPF)

Goal: Assess changes in 4DCT-derived lung compliance over time using linear regression

Hypothesis: we can observe significant decrease ($p < 0.05$) in lung compliance over time

- **Lung compliance**: measures the **distensibility** of the lung

$$\boxed{\mathrm{LC} = \frac{\Delta V}{\Delta P}}$$

Mechanical ventilation Commonly used	Intrapleural pressure measurement	Esophageal manometry	Four-dimensional CT (4DCT) Least invasive
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Math & Numerical Method

- Lung compliance

$$\boxed{\mathrm{LC} = \frac{\Delta V}{\Delta P}}$$

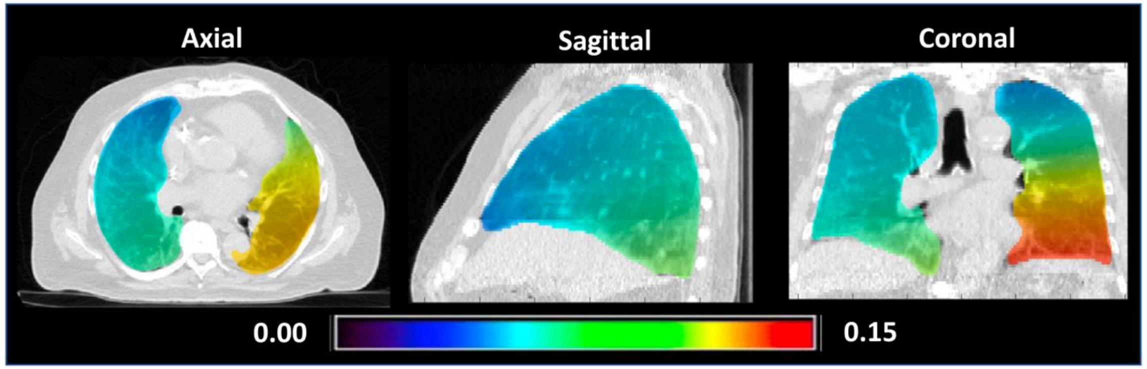
- 4DCT-Derived Lung compliance

$$\mathrm{vol}(\phi(\mathbf{x})) = J(\mathbf{x};\, \phi) \cdot \mathrm{vol}(\mathbf{x}).$$

$$LC(\mathbf{x}) = \frac{\mathrm{vol}(\mathbf{x}) - J(\mathbf{x};\, \phi) \cdot \mathrm{vol}(\mathbf{x})}{p_1 - p_2}.$$

$$C(\mathbf{x}) = 1 - J(\mathbf{x};\, \phi),$$

$$\boxed{C(x) = 1 - \frac{V_{P_1}}{V_{P_2}}.}$$



Lung compliance image of a IPF patient

4DCT scans with a segmentation mask

Nair et al., *Physics in Medicine & Biology*, 2021

Math & Numerical Method

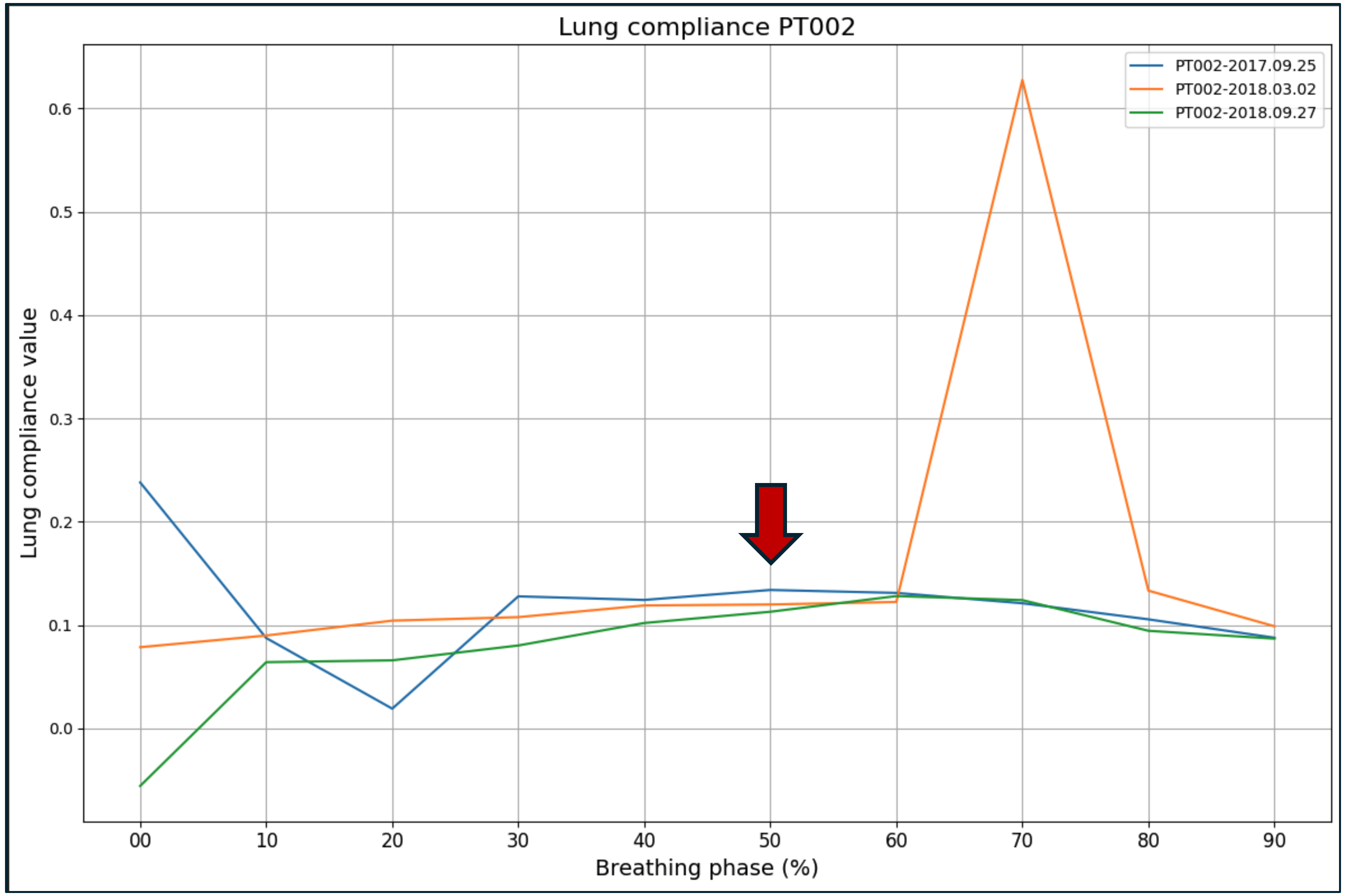
- 4DCT-Derived Lung compliance

$$\boxed{C(x) = 1 - \frac{V_{P_1}}{V_{P_2}}.}$$

- Dataset

  5 IPF cases, 3 timepoints

- For each case
  - Under 5, 10 cmH2O
  - Over breathing cycle (10 breathing phases)
  - Global lung compliance
  - At breathing phase 50

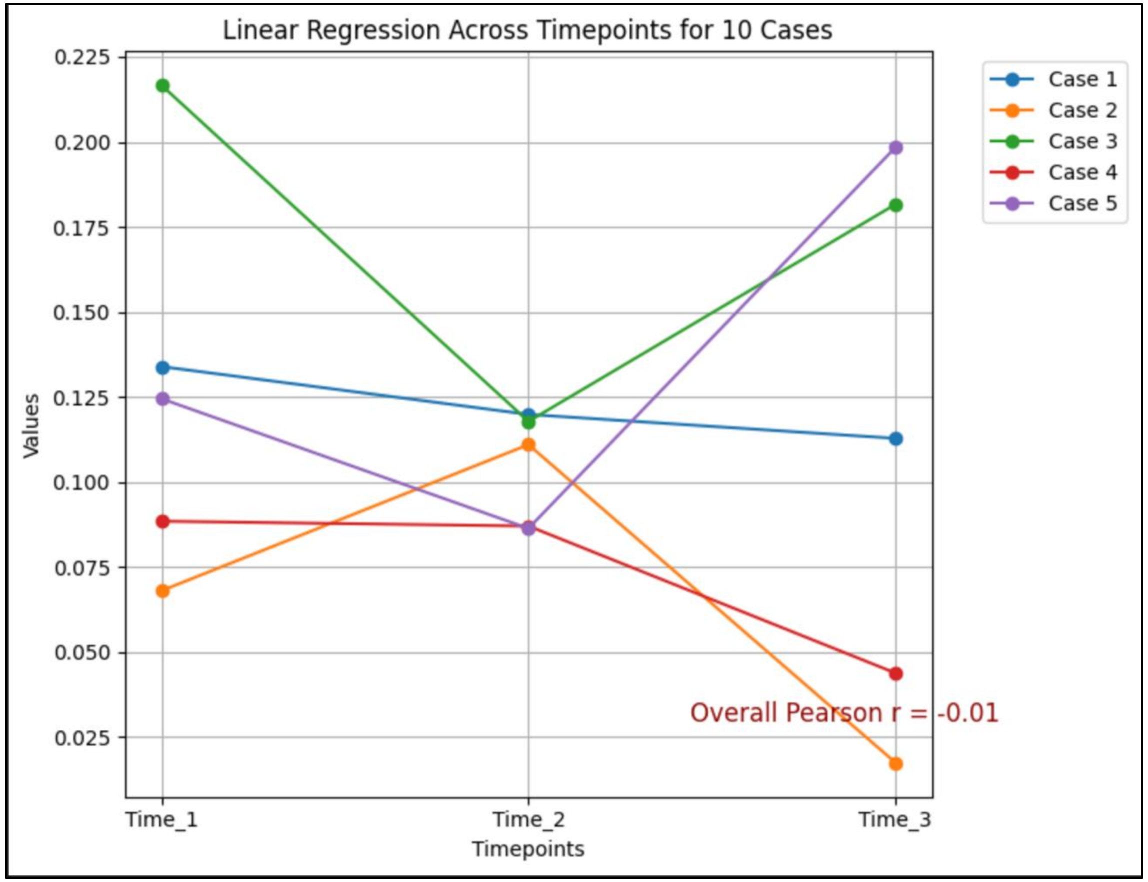


Lung compliance curve over the breathing cycle

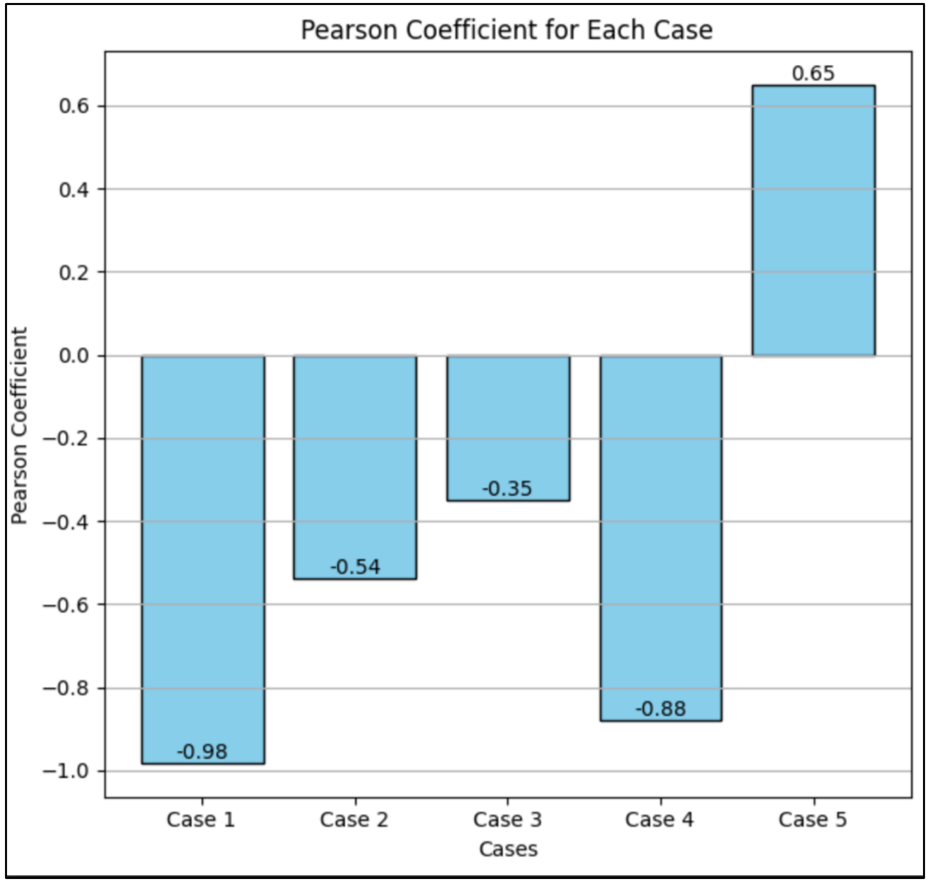
Math & Numerical Method

- Linear regression – Pearson correlation coefficient

$$r = \frac{\sum_{i=1}^{3}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{3}(x_i - \bar{x})^2 \sum_{i=1}^{3}(y_i - \bar{y})^2}}$$



Linear regression across 3 timepoints for each case



Pearson coefficient for each case

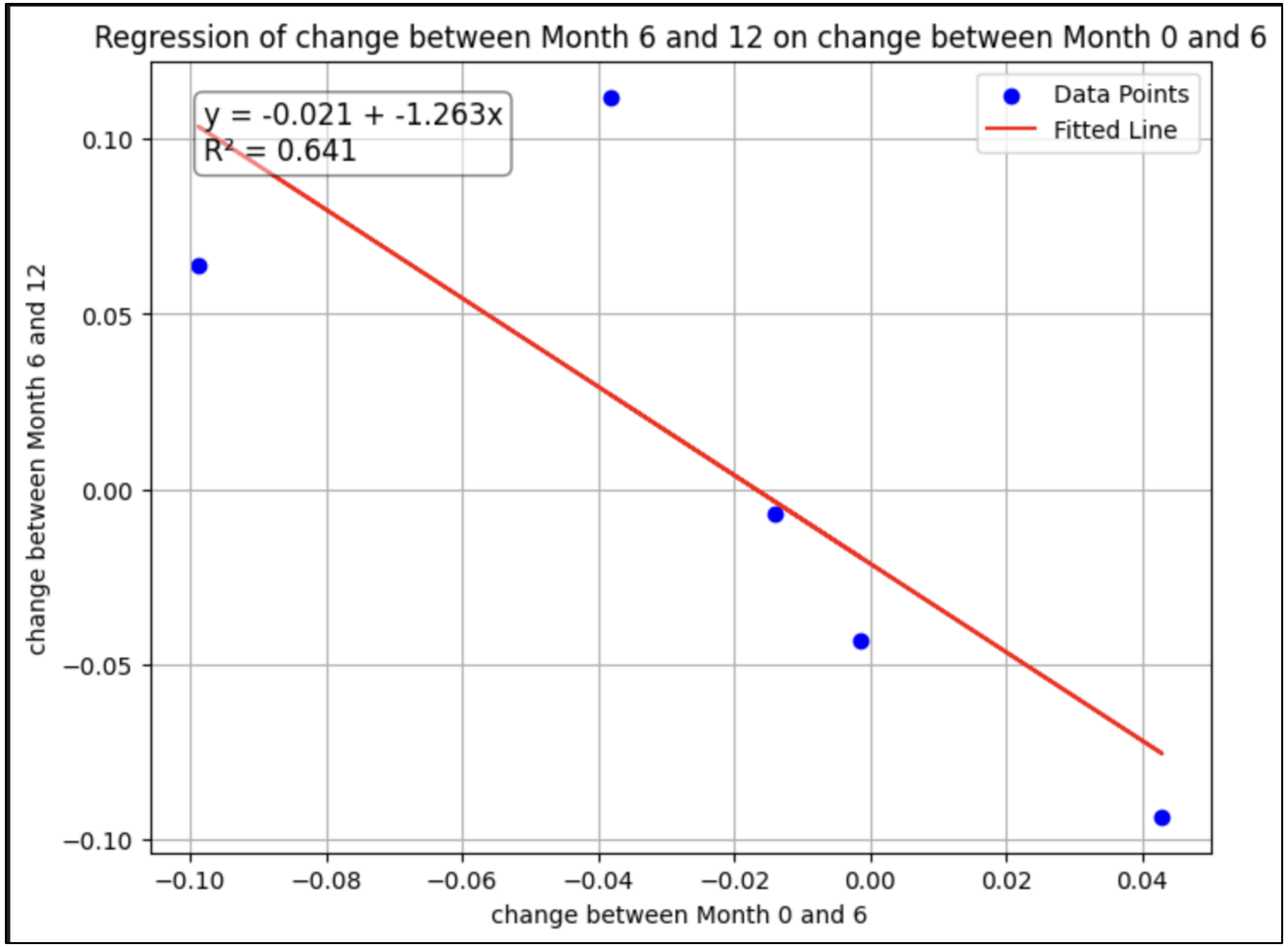
Math & Numerical Method

- Ordinary least square linear regression model

$$y = a_0 + a_1 x$$

$$a_1 = \frac{\sum_{i=1}^{5}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{5}(x_i - \bar{x})^2}$$

$$a_0 = \bar{y} - a_1 x$$




```python
def ols(y, X):
    y = np.array(y).reshape(-1, 1)
    X = np.array(X)

    beta = np.linalg.inv(X.T @ X) @ X.T @ y # find slope and intercept
    y_pred = X @ beta
    residuals = y - y_pred # Residuals: e = y - y_hat

    # R-squared: 1 - (SS_res / SS_tot)
    ss_res = np.sum(residuals**2)             # Sum of squared residuals
    ss_tot = np.sum((y - np.mean(y))**2)      # Total sum of squares
    r_squared = 1 - (ss_res / ss_tot)
    return beta.flatten(), y_pred.flatten(), residuals.flatten(), r_squared
```

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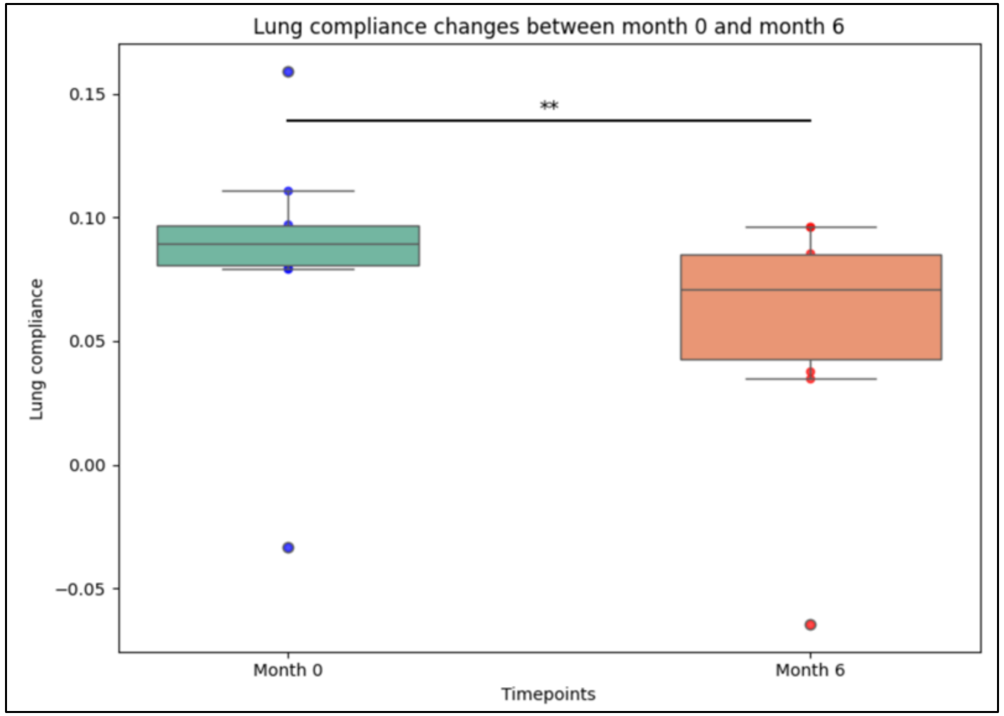
Summary and take-home messages

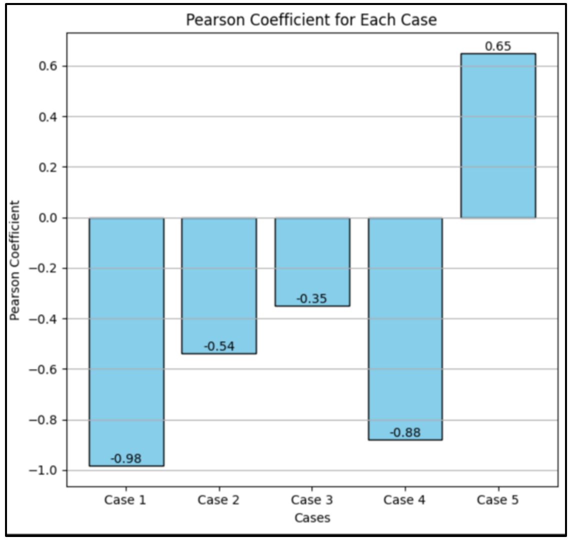
- Observed significant differences in lung compliance between different timepoints
- Regression across timepoints for each case (strongest r = -0.98)
- Relationship between changes from Month 0-6 and Month 6-12 (r = -0.8)

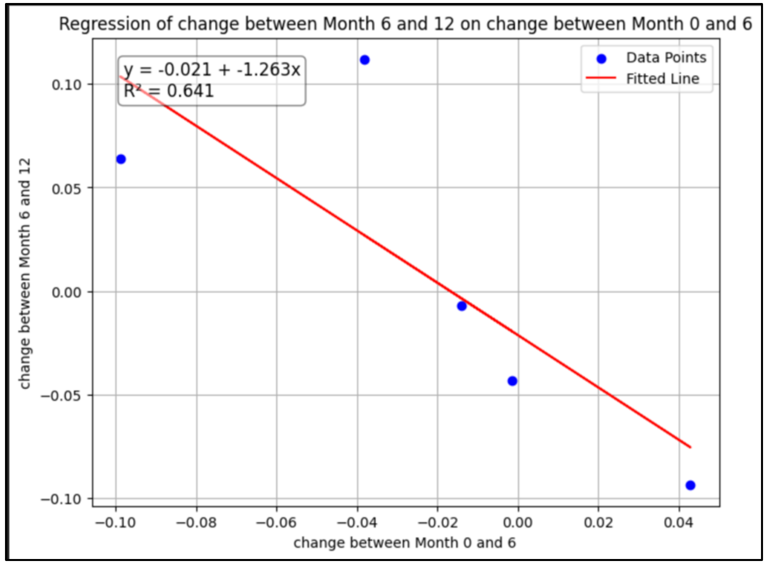


Changes in lung compliance between month 0 to 6. ** ($p < 0.01$)

Future direction

- Sine curve fitting remove outliers
- Collect more data
- Analyze regional lung compliance (in different lobes)





Thank you!

Any question?